AINETOps Reference Model & Problem Statement

2026-8-24

China Telecom, Deutsche Telekom,Cisco, China Unicom

Cisco Confidential

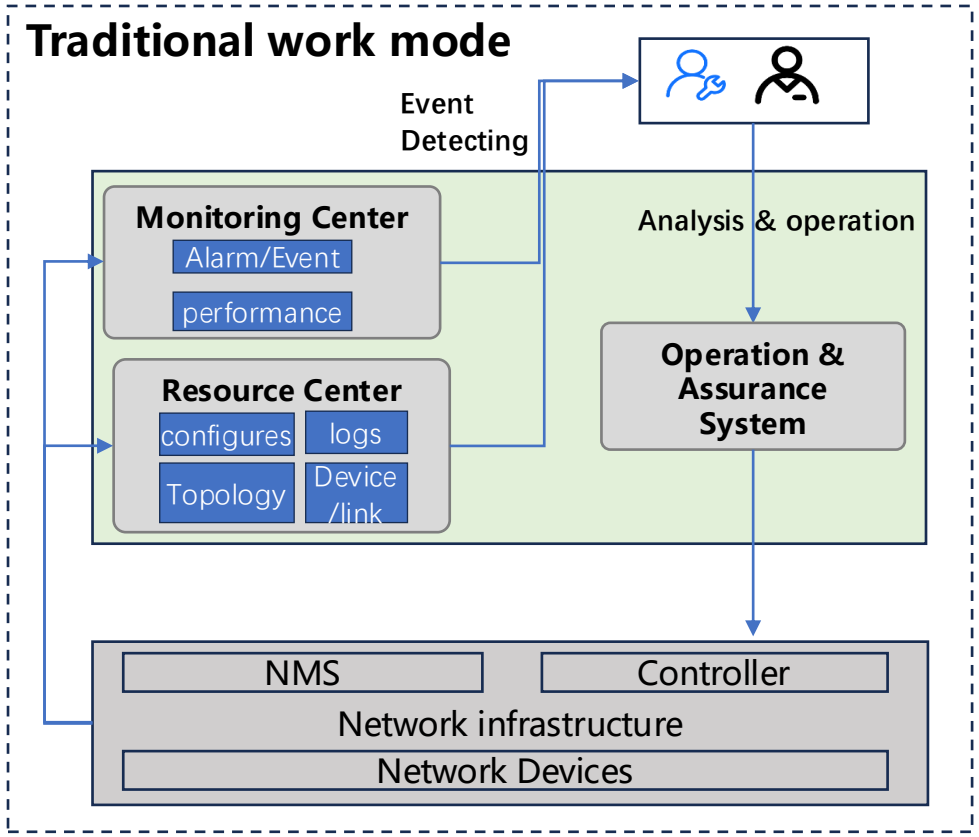
1 AINETOps Reference Model

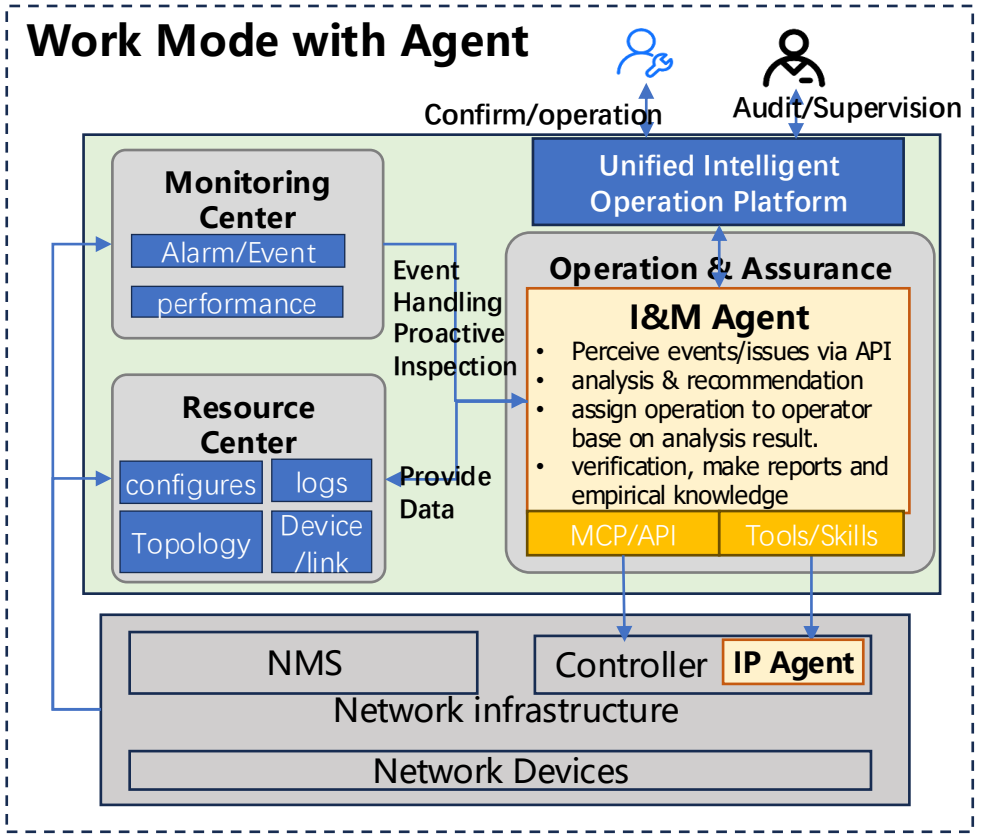
2 Problem statement for AINETOps

Cisco Confidential

Working mode shifts with AI agent

- **Human led,** OSS/NMS **passive** network management **->** **Based on intention,** human **monitoring and decision-making.**
- **Script+Static Workflow -> Self planning, predictability, intelligent reasoning**
- **Security is based on account/person -> based on task constraints supplement authenticated identities and authorization.**

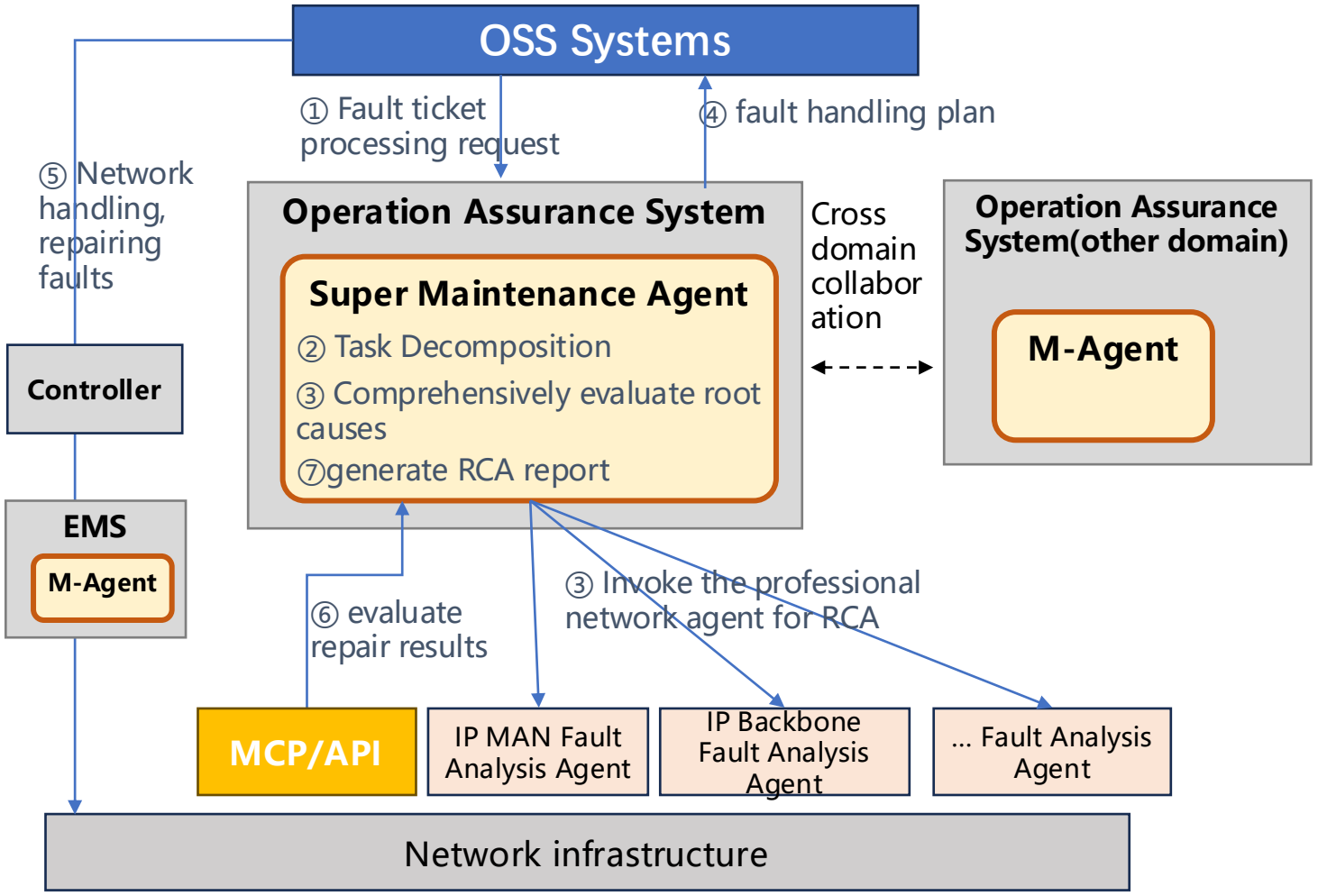




Cisco Confidential

Practice from China Telecom

Have applied 1000+ Network Management Agents and 8 major types of super agents in operational network, including network planning & building, service delivery, operation & assurance, network optimization, and integrated management.



Key Features:

① Achieves collaboration between t**he super-agent and single-function agents** across multiple domains

② Agents may be supplied by **different vendors**

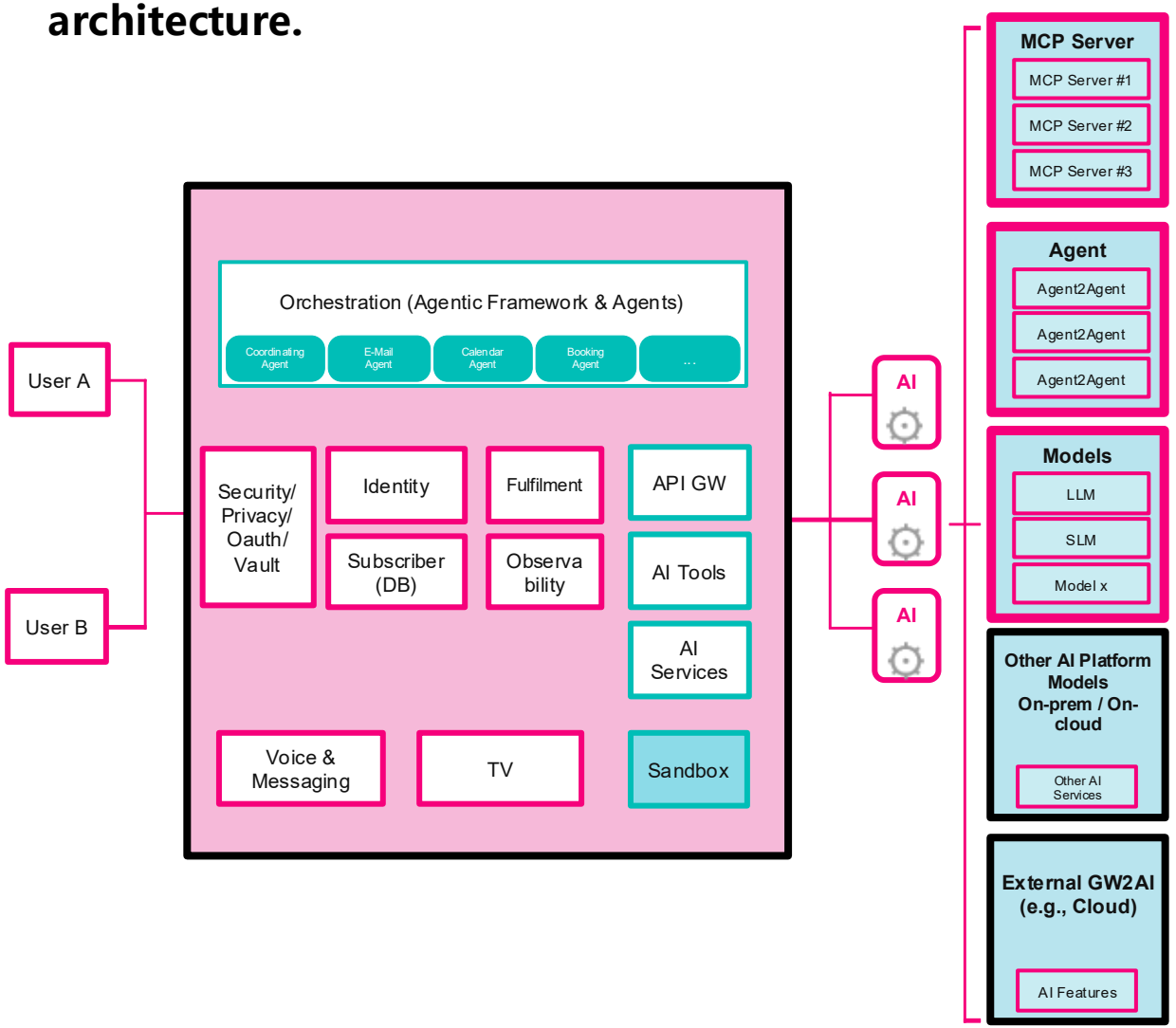
③ Agents are required to interact with multiple interfaces of legacy systems

④ Realizes end-to-end fault detection and root-cause analysis (RCA)

Cisco Confidential

Practice from Deutsche Telekom

Based on Agentic Orchestration as the core, it uniformly integrates Agents, models, MCPs and AI Services, and collaborates with traditional Telcom capabilities to form a standardized AI service connection and orchestration architecture.



- **Unified Access:** Connect Agents, models, tools, and internal/external AI services through the AI Gateway.
- **Unified Operations:** Provide management, monitoring, traffic control, cost control, security, and A2A/MCP capabilities.
- **Secure Launch:** Ensure the smooth operation of Agents through sandboxing, isolation, compliance checks, and lifecycle management.

Unify AgentOps governance to **support** plug-and-play functionality of Agents and large-scale operations.

Cisco Confidential

The Scope of AINETOps

AINETOPS is the place to find whether there is concrete work that falls under the IETF activity scope to cover AI integration, managing network management agents in network operations and their interaction with existing operational systems.

In Scope:

The introduction of **AI Agents** drives fundamental changes in O&M patterns and models, requiring a re-architected network operations framework, identifying the specific external boundaries under consideration.

- After introducing Agents into the operator’ s network O&M architecture, are **new interfaces** needed? Do existing interfaces require new **protocol support**?
- Define the **lifecycle management** framework for **Network Management Agents**.

Out of Scope:

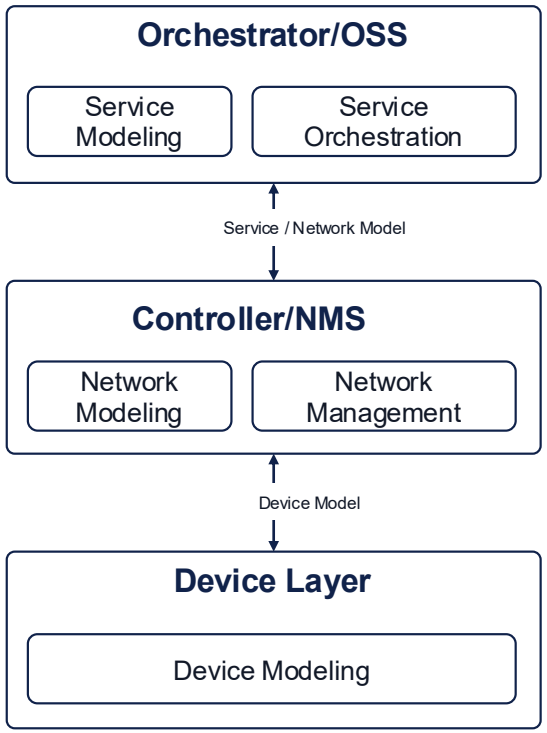
Discovery, Communication and Management of **General Agents**

Reference: Network Management Agent: draft-zhao-nmop-network-management-agent-05

Cisco Confidential

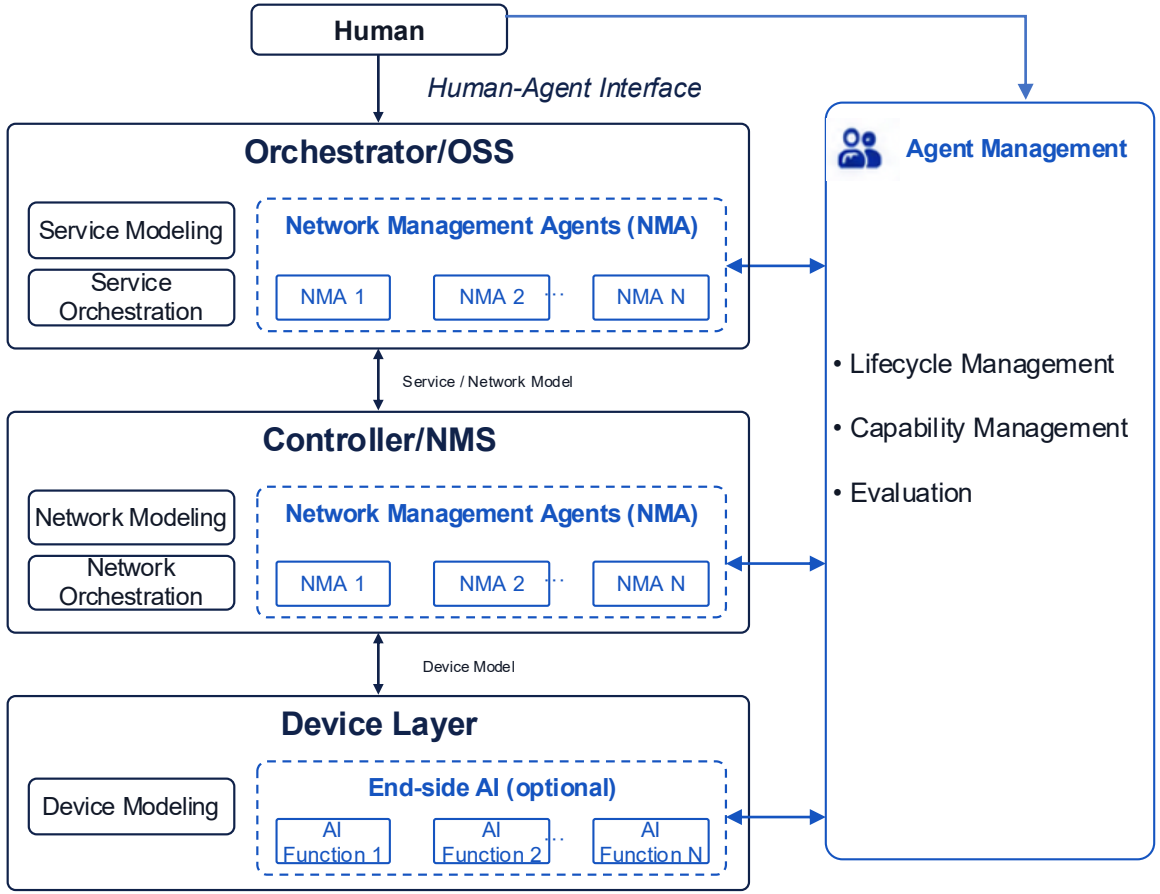
AINETOps Reference Model

IETF Reference Architecture



Extension for Agentic Networking

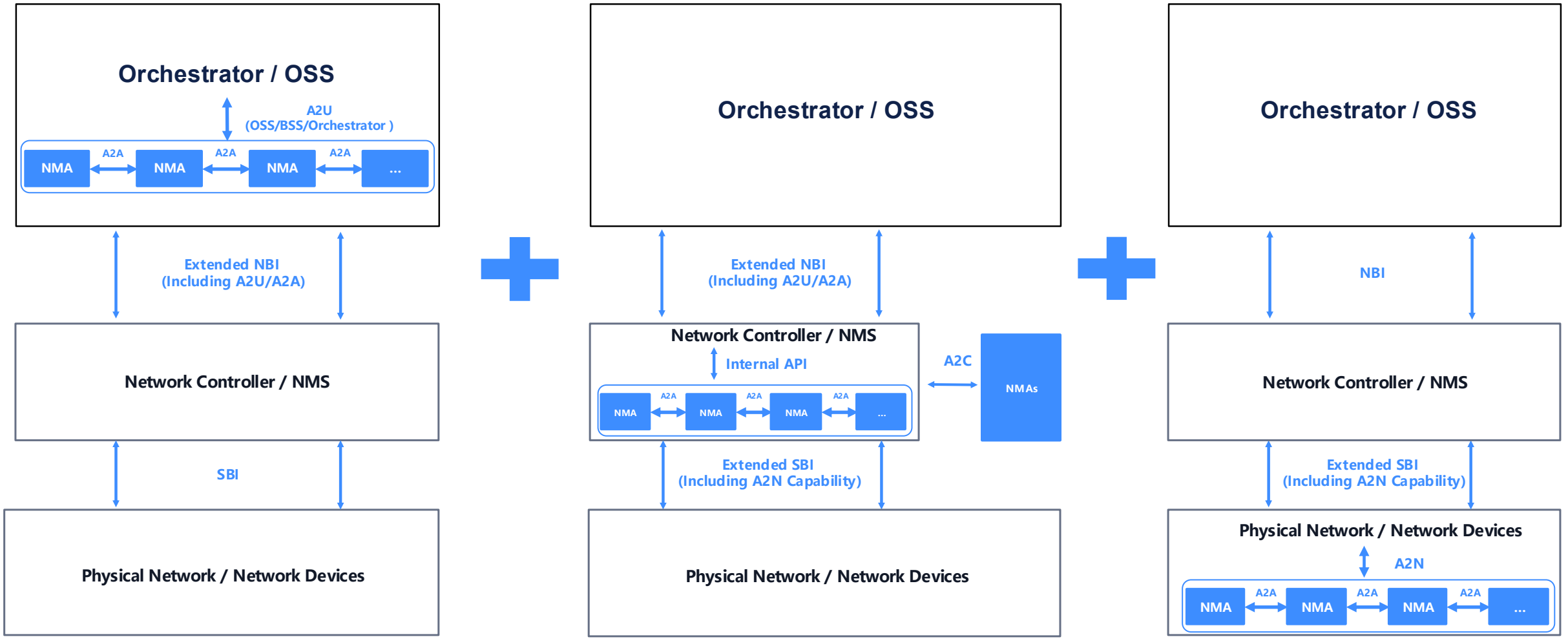
Extended Architecture with Agent



Cisco Confidential

Deployment mode of NMA

The three modes are: NMA deployed in the network management system, NMA deployed in the controller, and NMA deployed in the network devices.



Note: The three modes can be freely combined.

Cisco Confidential

1 AINETOps Reference Model

2 **Problem statement for AINETOps**

Cisco Confidential

Problem Statement initiated by AINETOps Reference Model

After Network O&M Model Restructuring

Why This Matters

The AINETOps Reference Model redefines network operations with AI-agents — but introduces critical standardization gaps in two dimensions:

- **P1: NMA Lifecycle Mgmt**
- **P2: End-to-End Multi-Vendor Agent Execute O&M Operations**

Problem 1

NMA Lifecycle Management

How do we manage the NMA (Network Management Agent) itself?

1) Framework: Standardized agent architecture & component boundaries
2) Agent Observability: Telemetry, tracing, state exposure for O&M personnel
3) Agent Evaluation / Evolution: Capability assessment, versioning, & continuous improvement

Problem 2

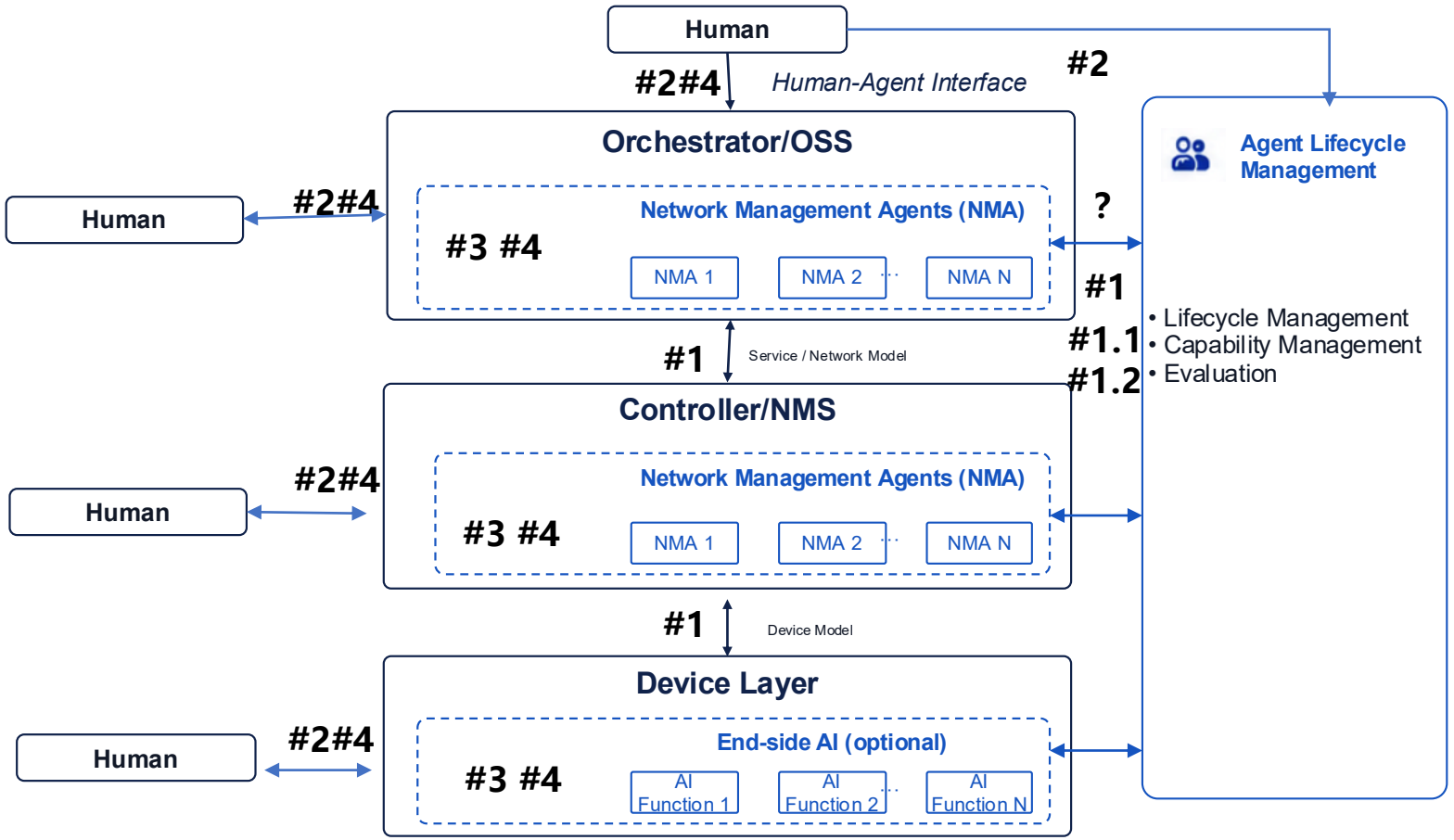
End-to-End Multi-Vendor Agent Execute O&M Operations

What new interfaces are needed? What information must they convey?

1) Human-Agent Interface (HAI): Human Intervention protocol & authority handoff semantics
2) NMA Task/Evidence Model (Agent-Agent Interface) : Is a standardized Task/evidence definition schema required?

Cisco Confidential

Problem : Interface Message Gaps



#1 NMA Lifecycle Management

#2#3#4 End-to-End Multi-Vendor Agent Execute O&M Operations : Trustworthy and Controllable

#1 Agent Lifecycle Mgmt → Problem 1

Unified Lifecycle Framework
Unified Observability Schema
Agent Telemetry & Metrics
Unified Evaluation approach
Unified mechanism to manage and verify the evolution of network operation

#2 Human-Agent Interface → Problem 2

Human–Agent Interface (HAI)

Operater ↔ Agent: intent input, stage confirmation, risk alert、breakoff、rollback

- **Agent-driven reporting:** execution results / status changes
- **Operator-driven push-down:** intent / correction policies / breakoff or rollback

#3 TASK Model → Problem 3

Agent-Agent Task Definition Schema

- **Task description:** intent / scope / constraints / status/outputs
- **Risk classification R0–R5:** permissions / approval / rollback

#4 Evidence Model → Problem 4

What / Where / When / Support

Agent ↔ Agent: Cross-domain evidence transfer: verifiable, traceable, and correlatable

- Identity (What) : evidence entries
- Quality / Traceability

Cisco Confidential

Problem 1: Agent Lifecycle Management—for Collection/Configuration

Unified Operational Lifecycle: Production Admission → Runtime Monitoring → Change / Rollback → Production Exit

1. Production Admission

- Capability declaration & constraints
- Operation scope & resource domains
- Maximum risk tier & operating mode
- Permission & policy verification

2. Runtime Monitoring

- Runtime status & health
- Task and operation execution
- Permission & resource monitoring
- Audit logs & alerts

3. Change / Rollback

- Configuration / policy / parameter updates
- Canary or limited-scope trial
- Rollback on change failure
- Archive change records & outcomes

4. Production Exit

- Revoke permissions & credentials
- Remove binding relationships
- Clear residual data & cache
- Archive logs & audit outcomes

Current Problems & Gaps

1 Fragmented State & Configuration

Across lifecycle stages, agent states and configurations are inconsistent. Information is difficult to reuse, weakening observability, auditability, and continuity.

2 No Versioned Evolution or Retirement

Models, knowledge, and tools lack a unified versioning and validation process. Vendor-specific exit logic creates orphaned access and compliance risks.

Existing Foundations (Across Working Groups)

IETF NMOP / NMRG	Network management agents, notifications, and policy work; covers capability discovery and model semantics.
TM Forum	A2A-T / Task-T task and interaction semantics, plus lifecycle management specifications.
ETSI / 3GPP	Network automation, configuration/change management, and security governance processes and standards.

AI NetOps Focus Areas (Theme Scope)

Unified Lifecycle Model

Define standard identities, registration, discovery, capability metadata, and trust establishment for network management agents—supporting unified multi-vendor governance.

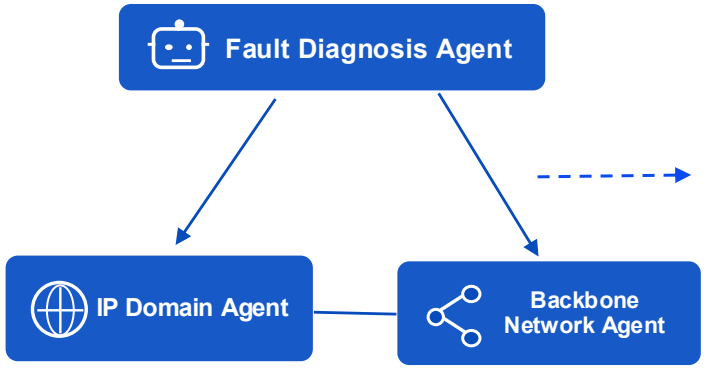
Versioned Change, Rollback & Offboarding

Define versioned mechanisms for capability and configuration changes, plus rollback, offboarding, and cleanup—ensuring end-to-end traceability and auditability.

Cisco Confidential

Problem 1.1: Agent Observability

Observability Example: Cross-Agent Task Execution Tracing



The observability of the Agent

Number of Agent calls	Abnormal Task Statistics
Task success rate	Rate Limiting Statistics
Agent connection status	Risk Type Statistics
Task execution time	.....

1 Lack of End-to-End Task Perspective Observability

Without a cross-component, end-to-end task view, it is impossible to tell which part — Agent, Tool, or other systems — is responsible for latency or failure. Task chains spanning multiple agents and tools become opaque black boxes.

2 Lack of Agent Perspective Observability

Without observability, it is difficult to know the connection status, model/agent/tool call logging, rate limit statistics, token consumption and API rate limits. Individual agent health and performance remain invisible to operations personnel.

Existing Foundation

IPFIX / Syslog — Standardized flow and event data collection (RFC 7011 / RFC 5424) provide infrastructure, but lack agent-specific semantic models.

NETCONF Notifications — Subscription-based notification mechanisms exist (RFC 5277), but do not cover agent internal state and model inference metrics.

AI NetOps – Topics to Advance

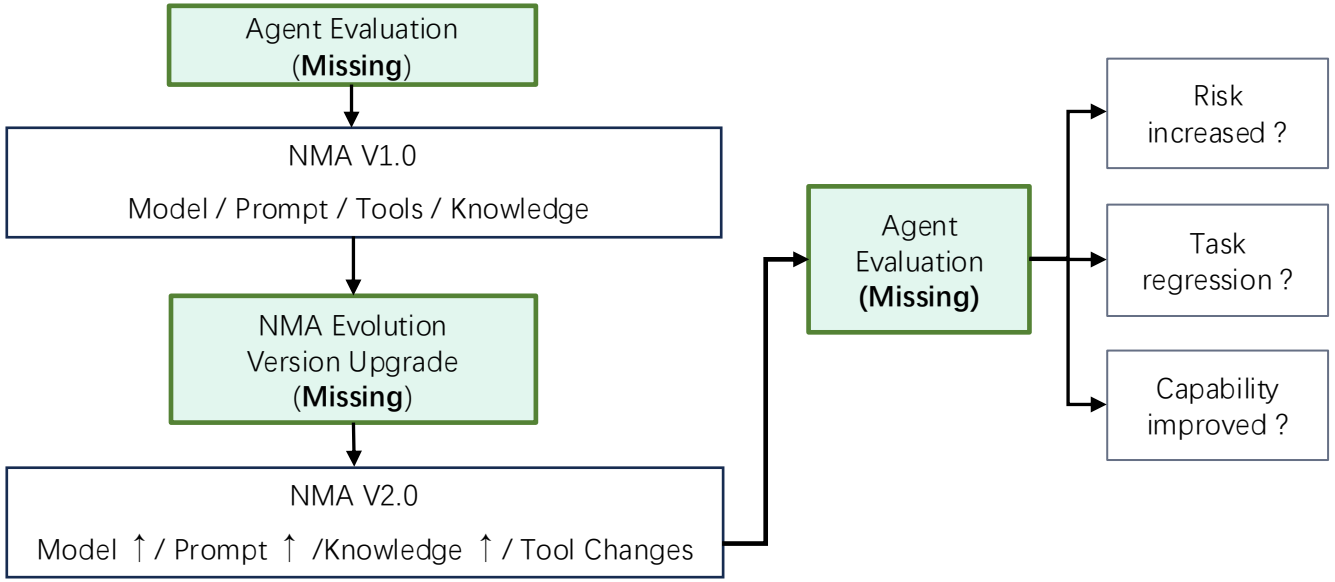
Unified Observability Schema — Define standard telemetry schema, performance metrics, and trace formats for cross-component task observability across agents and tools.

Agent Telemetry & Metrics — Define connection status, model/agent/tool call logging, token consumption, and API rate limit monitoring standards for agent operations.

Cisco Confidential

Problem 1.2: Agent Evaluation & Evolution

Evaluation & Evolution Example: Evaluation of the entire process of agent evolution



1 Agent Capabilities Lack a Unified Evaluation Baseline

- Network operation agents lack unified **standards for capability definitions**, **test tasks**, **datasets**, and **evaluation metrics**, **making objective and comparable evaluation difficult.**
- **Impact**: It is difficult to determine whether the agent meets the **production requirements**, and it is also difficult to objectively compare the capability differences among different agents and versions.

2 Agent Evolution Lacks Impact Validation and Regression Mechanisms

- Agent evolution lacks a unified mechanism to compare **new and previous versions** and identify **capability gains or regressions** across operational tasks.
- **Impact**: Unverified capability regressions may enter production, **increasing operational risks.**

Existing Foundation

Evaluation	BMWG is advancing Agent Benchmarking. There is still a lack of standards for evaluating intelligent agents.
Evolution	The IETF has not yet initiated any related work on the NMA evolution.

AI NetOps – Topics to Advance

Unified assessment method	Define what to evaluate, how to test, how to measure, and how to determine whether an agent meets requirements.
Agent evolution	Define triggering conditions, evolution scope, change mechanisms, and verification processes to ensure continuous capability improvement and operational reliability.

Cisco Confidential

Problem 2: Human-Agent Interface (HAI)

Unified Human-Agent Interaction Example: High-Risk Configuration Change Approval

Ops Personnel (Operator)

```
role: senior_engineer
intent: approve_change
target: BGP_Policy_Update
risk_level: R4
actions:
  - review_proposal
  - request_simulation
  - approve_with_conditions
constraints:
  - window: 02:00-04:00
  - rollback_ready: true
emergency_control:
  - one_click_rollback: on
  - force_stop: enabled
priority: critical
```

Human-Agent Interaction



Network Management Agent (NMA)

Order ID: 1234_XX
JobNo:
Status: 0(finish)
Analysis: Device 5 link 1 packet loss rate exceeding 5%, causing core network service call interruption rate exceeding 20%
Operator:
Option1: switch to backup tunnel
Risk: R2
Impact: VPN1 traffic
Simulation: required
Options2: shutdown dev5 link 1 。
Risk: R4
Impact: VPN1 traffic
Simulation: not required
Option3: no operation
Operator time: immediate/delay x hour
Evidence: *evidence info*

1 Loss Emergency Breakoff/Rollback Control

Agents lack phased confirmation mechanisms after launching tasks. When operators need to intervene, context has already been lost.
Operators cannot rapidly issue emergency rollback/isolation commands through an agent interaction.

2 Insufficient Option Detail& No Evidence Chain Attached

Each remediation option is presented with no structured **risk classification**, **impact scope**, or **simulation requirement**..
Agents lack phased confirmation mechanisms after launching tasks.
The analysis carries no attached logs, metrics, time window, or topology evidence.
Agents deliver conclusions without showing the supporting data trail.

Existing Foundation

CLI / Web GUI / Trap	Existing human-network interfaces provide basic command and alert channels, but lack structured semantics for agent interaction.
Syslog / Notification	Standardized event logging and notification formats exist (RFC 5424), but do not cover bidirectional human-agent dialogue.

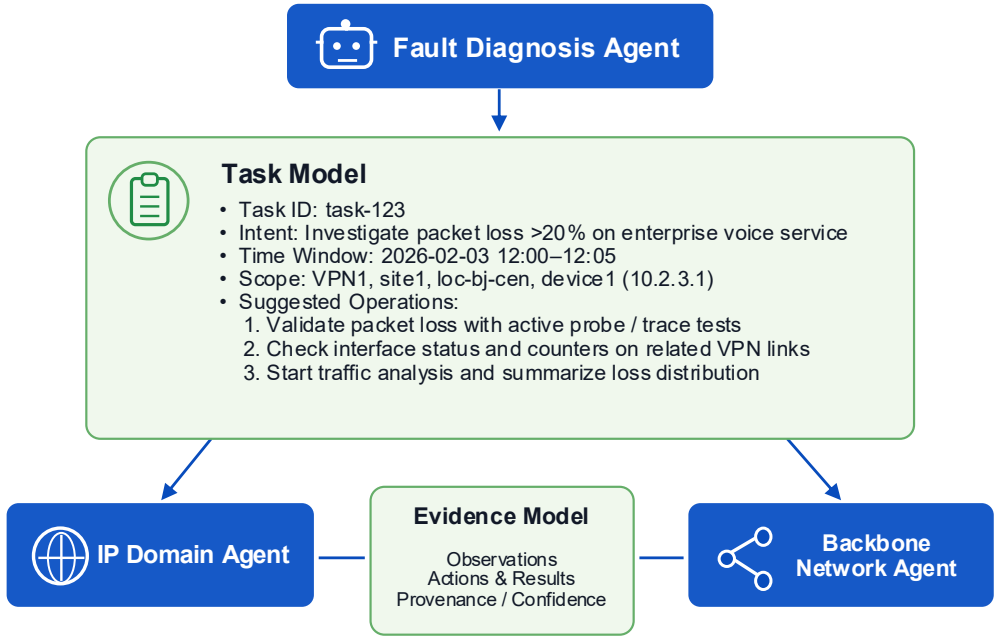
HAI– Topics to Advance

Human-Agent Interaction	Define structured message formats remediation options that includes action label, risk level, impact scope, affected services, and whether simulation is required before confirmation.
Evidence Chain & Auditability	Attach supporting evidence (logs, metrics, topology snapshots, time windows) to agent analysis, options, and operator decisions for real-time judgment and post-incident review.

Cisco Confidential

Problem 3&4: AI NetOps Task Model & Evidence Model

Unified Task Delivery Example: Congestion Alleviation for IP Transport



1 Lack of Unified Task Model among Agents

Across agents, controllers and domain tools, the same operation task is often expressed differently in goals, scope, constraints, execution bounds, status and outputs. This makes consistent delivery, delegation and task continuation difficult.

2 Lack of Unified Evidence Model

Observations, commands, intermediate results and verification data are reported in inconsistent formats and often lack shared provenance, confidence and trace context. This weakens cross-agent reasoning, auditability and human handoff.

3 Lack of Unified Risk Classification Standard

Production network tasks carry very different levels of risk, but there is no unified standard mapping risk levels to permissions, approvals, execution scope, rollback and verification requirements.

TM Forum Existing Foundation

A2A-T / Task-T	Defines task description, interaction flow and status management between agents.
Task-T / Event-T / Negotiation-T	Aligns with HVS, SID and shared telecom semantics to improve consistency of task meaning across agents.

AI NetOps – Topics to Advance

Evidence Model	Define evidence package, provenance, confidence, trace context and verification results for cross-agent reasoning and human review.
Task Risk Classification	Define R0–R5 risk levels and common assessment rules for network operation tasks.
Control Requirements by Risk Level	Define permissions, approvals, execution scope, rollback and verification requirements for each risk level.

Cisco Confidential

Thank you!

Cisco Confidential